FMCG SALES ANALYSIS & DEMAND FORECASTING

Predictive Analytics Project

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Executive Summary

This project analyzes Fast-Moving Consumer Goods (FMCG) sales data covering the period from 21 January 2022 to 31 December 2024. The objective is to understand sales patterns, investigate relationships between operational and promotional variables and units sold, and apply time-series techniques to examine aggregate daily sales and generate forecasts.

The analysis was conducted using Python, with Pandas and NumPy for data manipulation, Matplotlib and Seaborn for visualization, and Statsmodels for statistical modelling and forecasting.

The original dataset contained 190,757 records and 14 columns. Following data cleaning, 190,754 records were retained. Exploratory Data Analysis (EDA) revealed a right-skewed distribution of units sold, with an average of approximately 19.92 units per record and a median of 18 units. The dataset was distributed almost equally across the three regions represented in the data: PL-North, PL-South, and PL-Central.

Correlation analysis showed positive associations between units sold and stock availability (0.58), promotional activity (0.50), and delivered quantity (0.39). An Ordinary Least Squares (OLS) regression model achieved an R-squared value of 0.592, indicating that the selected predictors explained approximately 59.2% of the variation in units sold within the fitted model.

Time-series analysis was used to examine changes in aggregate daily sales, calculate a 30-day moving average, and compare Simple Exponential Smoothing (SES) and Double Exponential Smoothing (DES) with a simple last-value forecasting baseline. On the final 90 observations, the baseline achieved the lowest Mean Absolute Error (MAE) and Root Mean Squared Error (RMSE). Consequently, it was selected for the final 90-day forecast.

Overall, the project demonstrates how exploratory analysis, regression, data visualization, and forecast evaluation can be combined to investigate sales patterns and support more informed inventory and demand-planning decisions.

1. Introduction

Fast-Moving Consumer Goods are products that are purchased frequently and generally move through retail and distribution channels at a high rate. Understanding sales patterns in

this sector is important for monitoring demand, planning inventory, evaluating promotional activity, and coordinating replenishment.

Sales performance may be associated with several operational and commercial variables, including unit price, promotion activity, stock availability, delivery time, and delivered quantity. Analyzing these variables can help identify patterns in sales data and provide a basis for further investigation.

This project uses an FMCG dataset to explore the distribution of sales, examine relationships between selected variables and units sold, and study aggregate daily sales over time. The project also applies regression and time-series techniques to evaluate relationships and compare forecasting approaches.

The analysis focuses on the information available in the supplied dataset. The findings are observational and should not be interpreted as proof that any individual variable directly causes an increase or decrease in sales.

***Project Rationale**

This project was undertaken to apply business analytics techniques to a practical business context where sales, inventory, promotional activity, and delivery-related variables can be examined together. The FMCG sector provides a relevant setting for exploring these relationships and understanding how historical sales data can support demand planning. The project combines statistical analysis and forecasting to demonstrate how raw business data can be translated into meaningful insights for decision-making.

2. Problem Statement

FMCG sales data contains information about products, brands, regions, sales channels, promotional activity, inventory availability, deliveries, and units sold. Without structured analysis, it can be difficult to identify the relationships and trends contained within these records.

The project addresses the following questions:

1. How are sales records distributed across the regions represented in the dataset?
2. What is the distribution of units sold, and how much variation exists across records?
3. Which operational and promotional variables are most strongly associated with units sold?
4. How much variation in units sold can be explained by a linear regression model using the selected predictors?
5. How has aggregate daily sales changed during the observation period?

6. How do SES and DES compare with a simple baseline when forecasting the final 90 observations?

The analysis aims to answer these questions using descriptive statistics, visualizations, correlation analysis, OLS regression, and time-series forecasting.

3. Project Objectives

The primary objectives of this project are:

- To examine and prepare the FMCG dataset for analysis.
- To summarize the distribution of sales and operational variables.
- To visualize regional record counts and the distribution of units sold.
- To measure the relationships between numerical variables using correlation analysis.
- To fit an OLS regression model to examine the relationship between selected predictors and units sold.
- To explore historical sales trends using daily aggregation and a 30-day moving average.
- To compare forecasting methods using MAE and RMSE on a held-out test period.
- To generate a 90-day forecast and discuss the practical implications and limitations of the results.

4. Dataset Overview

The analysis uses the file FMCG_2022_2024.csv.

The dataset contains daily FMCG sales records covering multiple products, brands, segments, categories, channels, regions, and pack types.

4.1 Dataset characteristics

Attribute	Description
Original Records	190,757
Records after cleaning	190,754
Number of columns	14
Start date	21 January 2022
End date	31 December 2024
Regions	PL-North, PL-South, PL-Central
Target variable for regression	units_sold
Time-series measure	Aggregate daily units sold

4.2 Key variables

The dataset contains the following variables:

Variable	Description
Date	Date associated with the record
SKU	Stock-keeping unit identifier
Brand	Brand associated with the product
Segment	Product segment
Category	Product category
Channel	Sales channel
Region	Region associated with the record
Pack type	Product packaging type
Price unit	Unit price
Promotion flag	Indicator of promotional activity
Delivery days	Delivery duration in days
Stock available	Available stock
Delivered qty	Delivered quantity
Units Sold	Number of units sold, target variable for regression and measure used in time-series analysis

The analysis uses units_sold as the principal outcome variable. The explanatory variables selected for regression are unit price, promotion flag, delivery days, stock availability, and delivered quantity.

5. Data Cleaning and Preparation

Data cleaning was performed before the descriptive and statistical analyses to improve consistency in the data used for modelling.

The preparation process included the following steps:

Handling missing essential fields: Records missing units_sold, date, sku, brand, or region were excluded.

Date conversion: The date column was converted into a datetime format. Records with invalid dates were removed.

Filtering invalid sales values: Records with negative values of units_sold were excluded from the cleaned dataset.

Index reset: The cleaned dataset was assigned a fresh sequential index for subsequent analysis.

The original dataset contained 190,757 records, while the cleaned dataset contained 190,754 records. Therefore, three records were removed during the cleaning process.

Inspection of the original data showed that these three records also contained negative values in stock_available and delivered_qty. Since all three records also had negative units sold, they were excluded by the existing cleaning rule.

The resulting dataset contained 190,754 records and 14 columns and was used for descriptive statistics, EDA, correlation analysis, regression, and time-series analysis.

6. Methodology

The project was divided into four principal analytical stages.

6.1 Exploratory Data Analysis

Descriptive statistics were calculated to understand the central tendency, spread, and range of numerical variables. Visualizations were used to examine the distribution of units sold and the number of records associated with each region.

6.2 Correlation Analysis

A Pearson correlation matrix was calculated for price, promotion flag, delivery days, stock availability, delivered quantity, and units sold.

The matrix was used to identify the direction and strength of linear relationships between these variables. Correlation was treated as a measure of association rather than evidence of causation.

6.3 OLS Regression

An Ordinary Least Squares regression model was fitted using units_sold as the dependent variable and the following five explanatory variables:

- price_unit
- promotion_flag
- delivery_days
- stock_available
- delivered_qty

The model was used to examine how these variables were associated with units sold when considered together. R-squared, adjusted R-squared, coefficient estimates, p-values, and other regression summary statistics were used to interpret the results.

6.4 Time-Series Analysis and Forecasting

Units sold were aggregated by date to produce a daily time series. A 30-day moving average was calculated to smooth short-term fluctuations and make broader changes in sales easier to observe.

The project then applied two exponential smoothing techniques:

- **Simple Exponential Smoothing (SES):** Models the level of a time series.
- **Double Exponential Smoothing (DES):** Models the level and trend components of a time series.

For forecast evaluation, the final 90 observations were held out as the test period. The remaining observations were used to fit the models. Predictions were evaluated against actual test-period values using MAE and RMSE.

A simple last-value baseline was included for comparison. The method with the lowest test-period MAE was selected for the final 90-day forecast.

7. Results and Discussion

7.1 Descriptive Statistics

The cleaned dataset contained 190,754 records. The principal descriptive statistics for the numerical variables are presented below.

Variable	Mean	Median	Minimum	Maximum
Unit price	5.25	5.25	1.50	9.00
Promotion flag	0.149	0	0	1
Delivery days	3.00	3	1	5
Stock available	157.70	155	0	405
Delivered quantity	179.34	179	1	366
Units sold	19.92	18	0	139

The interquartile range for units sold was 12 to 25 units, meaning that the middle 50% of records fell within this interval.

The standard deviation of units sold was approximately 11.77, indicating substantial variation between records. The maximum observed value of 139 units was considerably higher than the median of 18 units.

The average promotion flag was approximately 0.149. Assuming a value of 1 indicates a promotion, approximately 14.9% of records were flagged as promotional.

These statistics describe the dataset as a whole. They do not independently establish differences in sales performance between individual products or regions.

7.2 Distribution of Records Across Regions

The regional record counts were:

Region	Records	Approximate share
PL-North	63,645	33.36%
PL-South	63,566	33.32%
PL-Central	63,543	33.31%
Total	190,754	100%

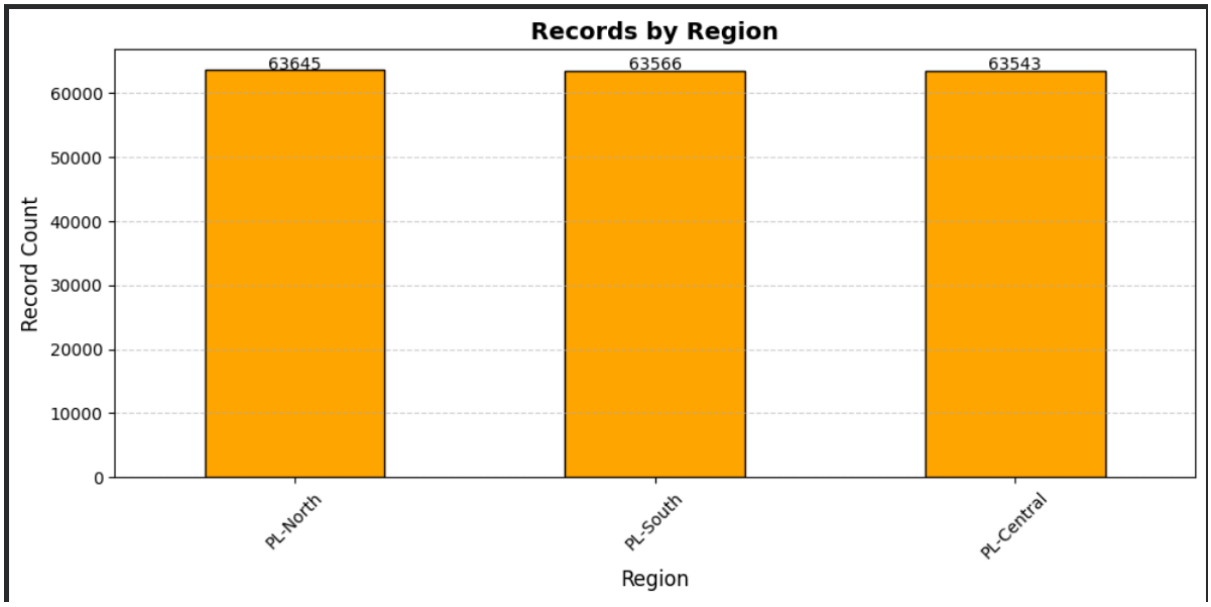


Figure 1: Distribution of Records Across Regions

The distribution of records was almost equal across the three regions. PL-North had the largest number of records, followed by PL-South and PL-Central.

The difference between the highest and lowest regional record counts was only 102. This indicates that the dataset's observations were distributed evenly across the three regions.

However, record counts should not be interpreted as regional sales rankings. A higher number of records does not necessarily imply higher sales volume, larger market size, or stronger regional demand.

7.3 Distribution of Units Sold

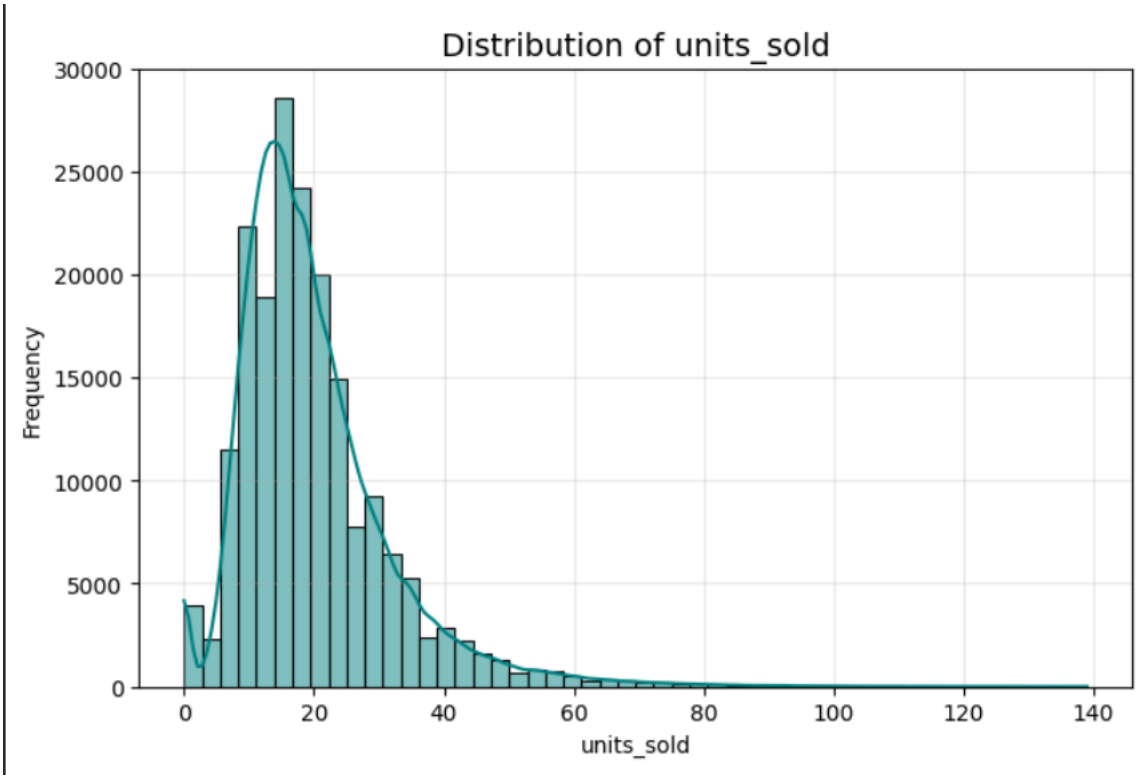


Figure 2: Distribution of Units Sold

The histogram of units sold showed a right-skewed distribution.

Most records were concentrated in the lower-to-moderate sales range, while a smaller number of observations extended towards substantially higher values. The mean of 19.92 units exceeded the median of 18 units, consistent with the right-skewed distribution.

The distribution suggests that sales volumes varied across records, with occasional high-volume observations. Such observations may be important when investigating product-level demand, promotions, or replenishment activity, although the histogram alone cannot establish the reasons behind individual peaks.

7.4 Correlation Analysis

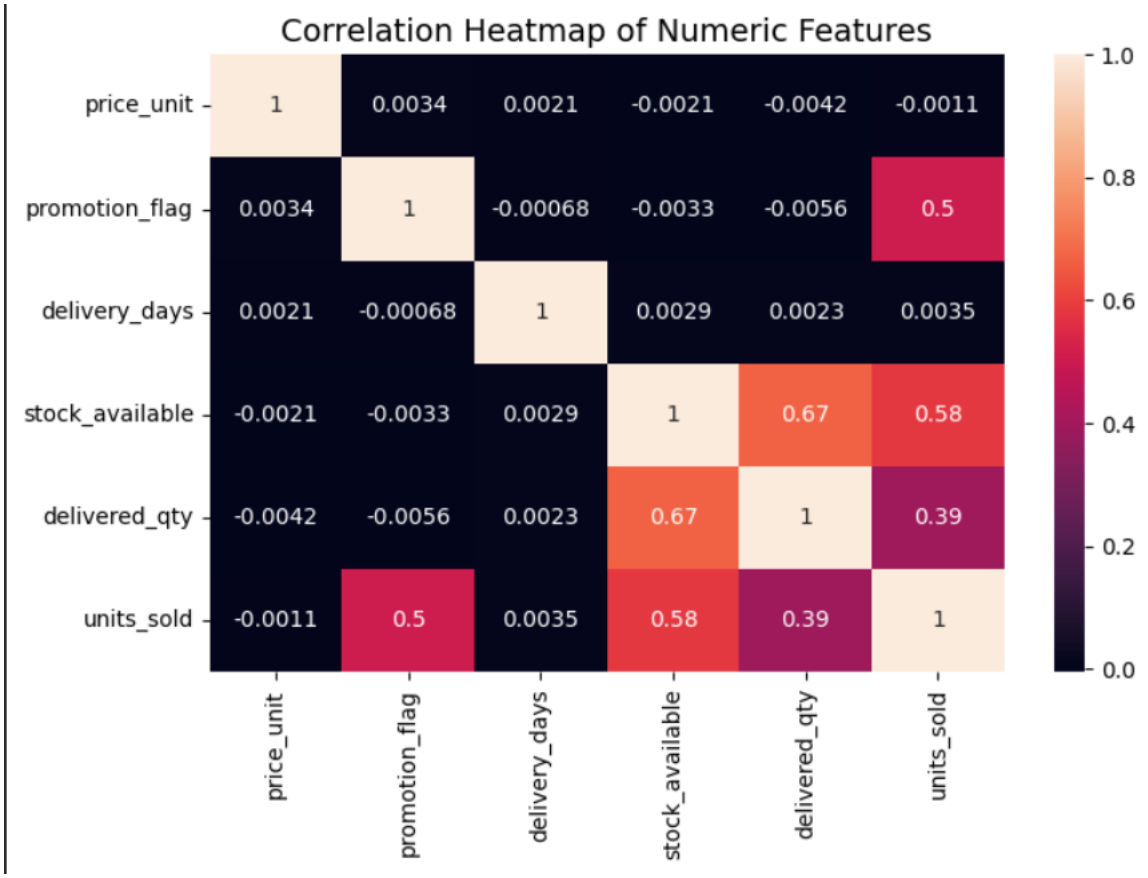


Figure 3: Correlation Matrix of FMCG Sales Variables

The correlation matrix produced the following principal relationships with units sold.

Variable	Correlation with units sold	Interpretation
Stock available	0.58	Moderate positive association
Promotion flag	0.50	Moderate positive association
Delivered quantity	0.39	Positive association
Delivery days	0.0035	Very weak linear association
Unit price	-0.0011	Very weak linear association

Stock availability had the strongest positive correlation with units sold among the variables examined, followed by promotion flag and delivered quantity.

The matrix also showed a correlation of approximately 0.67 between stock availability and delivered quantity. This indicates that the two inventory-related variables tended to move together in the dataset.

Price and delivery days had correlations close to zero with units sold. This indicates that these variables did not exhibit meaningful linear associations with sales at the overall dataset level in this analysis.

Correlation does not account for all possible relationships between variables. These results therefore identify associations for further investigation rather than proving that inventory or promotions directly cause higher sales.

7.5 OLS Regression Results

The OLS model used 190,754 observations and five explanatory variables.

The model achieved an R-squared value of 0.592 and an adjusted R-squared value of 0.592. This indicates that approximately 59.2% of the variation in units sold within the analysed sample was explained by the fitted linear model.

The model's overall F-test was statistically significant, with a reported p-value below 0.001. This indicates that the predictors, considered jointly, provided statistically significant explanatory information in the fitted model.

The coefficient results were as follows:

Predictor	Coefficient	P-value	Significance at 5%
Unit price	-0.0087	0.275	Not significant
Promotion flag	16.6864	Below 0.001	Significant
Delivery days	0.0178	0.143	Not significant
Stock available	0.1292	Below 0.001	Significant
Delivered quantity	0.0014	0.014	Significant

Promotion flag: The estimated coefficient was approximately 16.69. Holding the other included variables constant, records flagged as promotional were associated with approximately 16.69 more units sold than non-promotional records in the fitted model.

Stock available: The coefficient was approximately 0.1292. Within the model, an additional unit of available stock was associated with an increase of approximately 0.129 units sold, holding other predictors constant.

Delivered quantity: The coefficient was positive but relatively small, at approximately 0.0014. Its p-value of 0.014 indicated statistical significance at the 5% level.

Unit price and delivery days: Their p-values were 0.275 and 0.143, respectively. The model did not provide sufficient evidence to classify their coefficients as statistically significant at the 5% level.

These findings highlight the importance of examining promotional activity and inventory-related variables when studying sales variation. However, statistical significance does not prove a causal relationship. The results may also depend on omitted variables, model specification, and relationships among predictors.

The regression summary reported a Durbin-Watson statistic of approximately 1.299, residual skewness of 1.337, kurtosis of 9.864, and a condition number of approximately 1.39×10^3. These statistics indicate that the fitted model has potential diagnostic limitations, including non-normal residual behaviour and possible dependence or numerical issues. The findings should therefore be interpreted as exploratory rather than as a fully validated causal model.

7.6 Daily Sales Trends

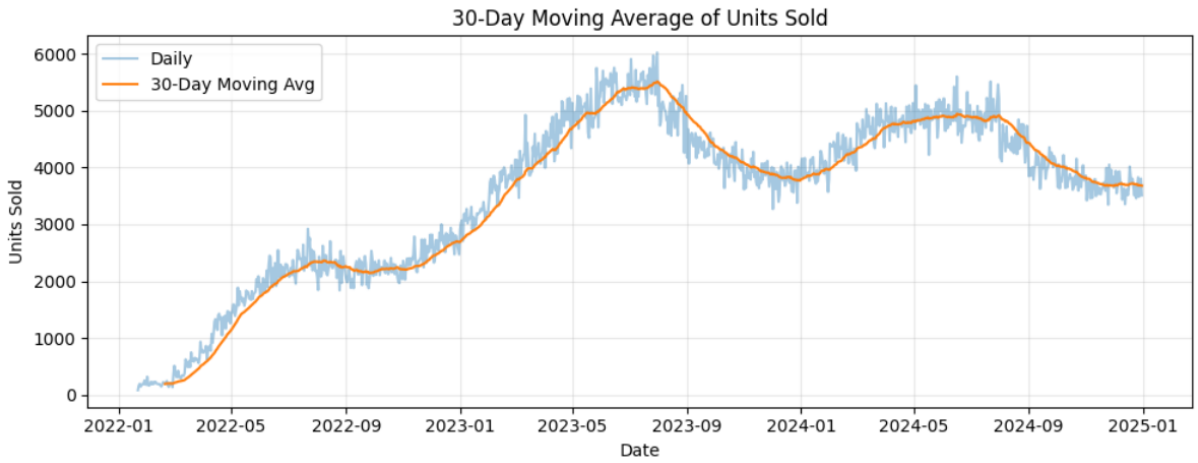


Figure 4: Daily Sales and 30-Day Moving Average

The time series was created by aggregating units sold across all records for each date.

The historical chart showed several broad changes in aggregate daily sales:

- Sales began at relatively low levels at the start of the observation period in January 2022.
- Sales increased substantially during the first half of 2022, followed by a period of comparatively stable levels later in the year.
- A pronounced increase occurred during the first half of 2023, with daily sales approaching approximately 5,500–6,000 units at peak levels.
- Sales declined during the latter part of 2023, reaching lower levels towards the end of the year.
- Sales recovered during the first half of 2024, reaching approximately 5,000 units on many days, before declining again towards the end of 2024.

These patterns indicate that aggregate daily sales did not follow a consistently upward trajectory throughout the entire period. The series contained periods of increase, decline, and recovery.

The 30-day moving average smoothed day-to-day fluctuations and made these broader movements easier to observe. It provided a useful descriptive view of the trend but did not independently identify the causes of changes in sales.

7.7 Forecast Evaluation: SES, DES, and Baseline

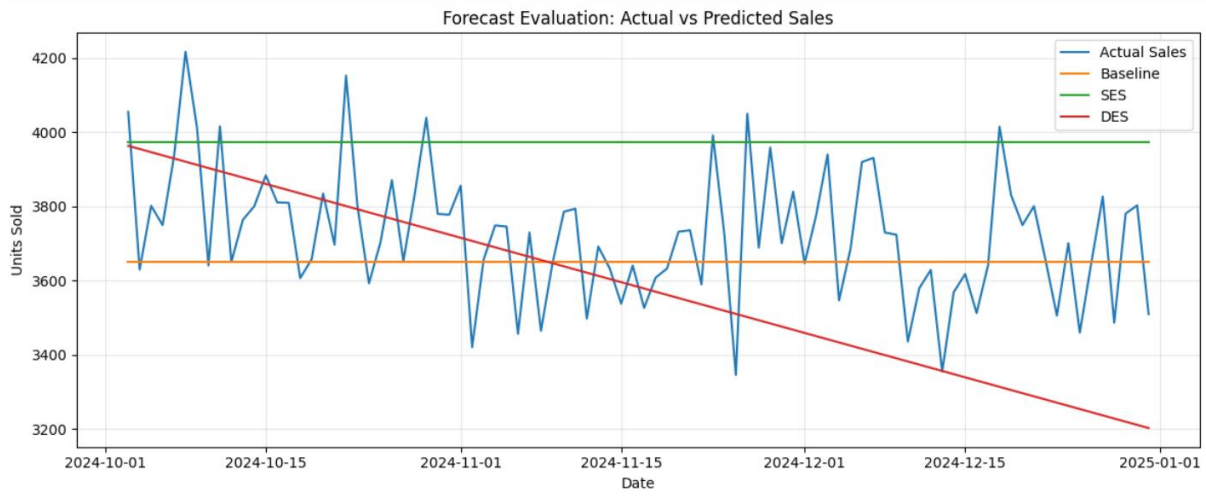


Figure 5: Forecast Evaluation — Actual vs. Predicted Sales

The forecasting evaluation compared three methods on the final 90 observations:

1. Last-value baseline
2. Simple Exponential Smoothing (SES)
3. Double Exponential Smoothing (DES)

The models were fitted using the earlier observations, and their predictions were compared with actual sales during the held-out test period.

Model	MAE	RMSE
Last-value baseline	144.97	188.58
DES	217.81	274.83
SES	265.23	302.49

MAE measures the average absolute difference between predicted and actual sales. RMSE also measures forecast errors but gives greater weight to larger errors. For both metrics, lower values indicate better performance on the same test period.

The last-value baseline achieved the lowest MAE and RMSE among the three methods. DES performed second, while SES had the highest errors.

The baseline's MAE of 144.97 was lower than DES's 217.81 and SES's 265.23. Its RMSE of 188.58 was also lower than the corresponding values for DES and SES.

This result is important because it demonstrates that a more complex forecasting technique is not necessarily more accurate than a simple benchmark. Under the current model settings and test period, the exponential smoothing methods did not outperform the last-value baseline.

The comparison is limited to one held-out period. It does not establish that the baseline will always be superior under different demand patterns or future conditions.

7.8 Final 90-Day Forecast

Following the forecast evaluation, the last-value baseline was selected because it had the lowest MAE.

The final forecast used the last observed aggregate daily sales value of 3,509 units as the prediction for each of the next 90 days after the dataset's final date.

The output covered the period from 1 January 2025 to 31 March 2025, with a constant forecast of 3,509 units per day.

This is a simple persistence forecast. It assumes that the latest observed sales level will continue and does not model an upward trend, downward trend, or seasonal pattern. It provides a benchmark estimate rather than a detailed operational demand forecast.

The forecast should therefore be interpreted cautiously. Its value in this project is that the method was selected using an explicit test-period comparison rather than assuming that SES or DES was automatically the best-performing model.

8. Business Implications and Recommendations

The analysis suggests several ways in which the findings could inform further FMCG sales and inventory analysis.

8.1 Monitor Inventory and Sales Together

Stock availability had a positive correlation of 0.58 with units sold and was a statistically significant predictor in the OLS model. Businesses could monitor inventory and sales together to identify products or regions where changes in availability coincide with changes in sales.

The results do not establish the optimal stock level. Further analysis at product and regional levels would be required before setting inventory targets.

8.2 Evaluate Promotional Activity

Promotion flag had a correlation of 0.50 with units sold and a positive, statistically significant coefficient in the regression model. These findings justify further examination of promotional performance.

A useful next step would be to compare sales for promotional and non-promotional records while accounting for product, channel, region, and time differences. This would help distinguish the observed association from the actual incremental effect of promotions.

8.3 Review Replenishment Patterns

Delivered quantity was positively associated with units sold and was statistically significant in the regression model. Since delivered quantity and stock availability were also correlated, inventory and replenishment decisions should be examined together.

Further analysis could assess whether deliveries occur before or after sales changes and whether the relationship differs across products or regions.

8.4 Track Sales Trends Over Time

The 30-day moving average offers a simple way to monitor broader changes in daily sales. Businesses could use comparable rolling indicators to investigate sustained increases or declines and determine when a more detailed review may be needed.

8.5 Validate Forecasting Models Before Operational Use

The last-value baseline outperformed SES and DES in the selected test period. This highlights the need to evaluate forecasting methods against actual unseen observations rather than choosing a model solely because it is more sophisticated.

Repeated time-based testing over multiple forecast windows could help determine whether the same model continues to perform well under different conditions.

9. Limitations

The findings should be interpreted in light of the following limitations.

Observational data: The analysis identifies associations but does not establish causal effects. For example, a positive relationship between promotions and sales does not by itself prove that promotions caused the observed increase.

Regression diagnostics: The OLS summary indicates potential issues involving residual behaviour and numerical stability. The current analysis does not include a complete regression-diagnostics workflow, robust standard-error analysis, or a formal model-validation exercise.

Aggregate time-series analysis: The forecasting analysis uses aggregate daily units sold across all records. It does not generate separate forecasts for individual SKUs, brands,

channels, or regions. Aggregation can hide differences in demand patterns across these groups.

Forecast evaluation period: SES, DES, and the baseline were compared on a single held-out period of 90 observations. Their relative performance may differ over another period.

Forecasting methods: The analysis uses simple smoothing approaches and a last-value baseline. It does not establish whether a seasonal model or a model incorporating additional explanatory variables would perform better.

Data provenance: The notebook does not document an external source for the supplied CSV file. The original dataset source should be acknowledged in the final public report or repository if it is known.

These limitations do not invalidate the exploratory findings, but they define the scope of the conclusions that can reasonably be drawn from the project.

10. Conclusion

This project combined data preparation, exploratory analysis, correlation analysis, OLS regression, moving averages, and time-series forecasting to examine FMCG sales patterns.

The cleaned dataset contained 190,754 records across three regions. Descriptive statistics showed that units sold had a mean of approximately 19.92 and a median of 18, with a right-skewed distribution. Regional record counts were almost evenly distributed across PL-North, PL-South, and PL-Central.

Correlation analysis identified positive relationships between units sold and stock availability, promotion flag, and delivered quantity. The OLS regression model achieved an R-squared value of 0.592, with promotion flag, stock availability, and delivered quantity statistically significant at the 5% level in the fitted model. Price and delivery days were not statistically significant at that threshold.

The historical daily sales series showed periods of increase, decline, and recovery. A 30-day moving average helped reveal these broader patterns while smoothing short-term fluctuations.

Finally, forecast evaluation showed that the last-value baseline outperformed SES and DES on the selected test period. The baseline was therefore selected for the 90-day forecast. This result reinforces the importance of objective model evaluation and demonstrates that simple forecasting approaches can provide a useful benchmark against more complex techniques.

Overall, the project illustrates how Python-based analytics can turn structured sales data into interpretable findings about sales variation, inventory-related measures, promotional activity, and historical demand. The results provide a foundation for further investigation

and more robust forecasting work, while acknowledging that additional validation would be required before applying the findings to operational decisions.